

# High-Efficiency Ultrawideband Microwave Rectifier Based on Adaptive Harmonic Control

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**Abstract**—Harmonic control is an efficient design method to improve the efficiency of a microwave rectifier. Conventional class  $F$  and  $F^{-1}$  microwave rectifiers achieve high efficiencies by harmonic control. However, the working bandwidth is limited since the harmonics generated during rectification have to be strictly controlled. This article presents a method to design a microwave rectifier that adaptively switches between class  $F$  and  $F^{-1}$  with respect to frequency variations by a harmonic control network. A broadband rectifier is designed with an efficiency higher than 70% from 1.54 to 2.66 GHz at 11-dBm input power. Extending the method to apply more harmonic termination switches, another ultrawideband rectifier is designed, fabricated, and measured. The ultrawideband rectifier maintains a rectification efficiency higher than 70% over the frequency range from 1.0 to 3.1 GHz. The design and measurement results of both rectifiers verify the high efficiency and effectiveness of the proposed adaptive method of harmonic control, which successfully breaks the frequency limitation of conventional harmonic terminal control. The design can be applied to wireless power transfer and energy harvesting.

**Index Terms**—Broadband, energy harvesting (EH), harmonic control, high-efficiency, rectifier, wireless power transfer.

## I. INTRODUCTION

MICROWAVE Wireless Power Transfer (MWPT) and energy harvesting (EH) technologies have demonstrated significant research importance and broad application prospects across various fields. [1], [2], [3] MWPT transfers energy from the transmitter to the receiver via microwaves, making it suitable for long-distance and high-power transmission applications such as wireless charging for electric vehicles, energy replenishment for satellites and space stations, and dynamic charging for drones. These applications not only enhance the convenience and safety of equipment but also reduce dependency on traditional cables and batteries, contributing to environmental protection [4], [5], [6]. EH technology, on the other hand, directs microwave sources toward surrounding RF devices, collecting electromagnetic energy from the environment and converting it into usable electrical power. This is particularly suitable for low-power devices such as wireless sensor nodes, wearable devices, and smart homes [7], [8]. EH technology can significantly

extend the operational lifespan of devices, reduce maintenance costs, and provide a continuous power supply in remote and specialized environments. A highly efficient and stable rectifier is essential for converting the received electromagnetic energy. The rectifier must efficiently convert energy across different frequencies, posing a challenge for its design.

Among various research efforts, enhancing rectifier efficiency remains a primary goal. Numerous solutions have been proposed and validated as effective design methods, promoting the advancement of wireless power transfer and space EH technologies [9]. Classical rectifier circuits typically use nonlinear devices such as Schottky diodes or GaN transistors [10] to rectify, filters to reduce clutter, and resistive loads to receive the dc power. Experimental results in [11] and [12] demonstrate that the rectifier circuit can achieve conversion efficiencies of 70% and 71%, respectively. Harmonic termination techniques are commonly used to improve the efficiency of power amplifiers (PAs), as the nonlinear characteristics of transistors generate harmonic components. To improve the efficiency of the amplifier, the terminations of the higher harmonics are set to change the waveforms of voltage and current, thereby enhancing overall efficiency. Roberg and Popovic [13] presented a general analysis method for arbitrary complex harmonic terminations, including a sensitivity analysis of the termination impedance.

In rectifiers, nonlinear rectifying elements not only generate a dc output but also generate higher harmonic components of the input frequency. The efficiency of a rectifier can be significantly improved through proper harmonic termination design. For example, a diode rectifier designed with the concept of class  $C$  harmonic termination is integrated with a patch antenna for wireless power transfer systems [14], significantly enhancing the rectifier's efficiency. Liu et al. [15] utilized a series band-stop structure to achieve harmonic recovery, achieving an efficiency of 80.9% at 2.45 GHz, and this method was also applied to a voltage-doubling rectifier design in [16]. The circuit topology of a class  $F$  rectifier incorporates a harmonic termination network in design, typically using open/short-circuit stub lines and a  $\lambda/4$  transmission line (TL). An analytical model is derived based on the diode voltage and current waveforms in the time domain to calculate the rectifier efficiency of class  $F$  rectifiers [17]. In [18], a class  $F$  rectifier implements a termination topology that eliminates the need for a conventional output filter. This design achieved 71% efficiency with 14.7-dBm input power. Zhao et al. [19] proposed a compact class  $F$  rectifier, achieving an 81.4% efficiency improvement at watt-level input power. Due to the stringent frequency requirements of harmonic terminations,

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this technology is currently mainly applied in single-frequency or dual-frequency rectifier designs.

The continuous class  $F$  concept used in PA design has been extended to rectifier design to apply the high-efficiency characteristics over a broader bandwidth. However, efficiencies greater than 50% are only achieved within the range of 0.78–1.43 GHz [20]. Research on high-efficiency wideband rectifiers primarily focuses on enhancing the operating bandwidth from the perspectives of frequency-selective diode arrays [21], broadband matching networks [22], [23], and impedance compression. For example, nonuniform TLs have been employed for impedance matching in broadband rectifiers, successfully achieving high rectification efficiency over a wide frequency range [24], [25]. Simplified real-frequency techniques [27] and complex impedance matching networks [28] have been successfully applied in wideband rectifier designs. Regarding novel impedance compression techniques, the key to achieving compact size and high efficiency is to reduce the variation range of the rectifier's input impedance [29], [30], [31], [33]. The stability of the impedance of the double voltage structure helps to expand the bandwidth of the rectifier [34], [35], [36], [37]. The maximum score bandwidth can reach 197.6% in [37]. A major challenge in rectifier design is how to expand the operating bandwidth while maintaining high efficiency.

Harmonic termination control is an effective method for improving efficiency. However, the strict frequency requirements limit the application of harmonic termination techniques in wideband rectifiers, reducing the efficiency of such devices. Continuous class  $F$  and  $F^{-1}$  theories suggest that strict open-circuit and short-circuit conditions are not the only way to maintain high efficiency in PAs [38], and the same principles apply to rectifiers. Therefore, we propose a high-efficiency wideband rectifier design scheme based on a harmonic control network, which allows the rectifier's harmonic termination mode to vary with frequency. The design switches from a standard class  $F^{-1}$  to class  $F$ , thus ensuring the high efficiency achieved through harmonic termination control while also expanding the rectifier's operating bandwidth. Compared with our previous work [39], this article provides a more in-depth analysis and explanation of the principles of harmonic control networks. The theory is also further extended, resulting in a further increase in the operating bandwidth of the rectifier. The expanded harmonic control network can transform multiple harmonics within the operating bandwidth, which makes the harmonic control network more widely adaptable. This article analyzes and extends the theory, and two high-efficiency wideband rectifiers were fabricated and tested, verifying the feasibility of the theory. The highest efficiency, exceeding 70%, was achieved over more than 100% fractional bandwidth.

## II. HARMONIC-CONTROL RECTIFIER THEORY

### A. Standard Class- $F/F^{-1}$ Modes

In conventional microwave rectifier design, using nonlinear diodes for half-wave rectification generates additional harmonic power. The generated nonzero harmonics are one of the key factors leading to decreased rectifier efficiency. To effectively address this challenge, researchers have developed various harmonic termination techniques. Among them,

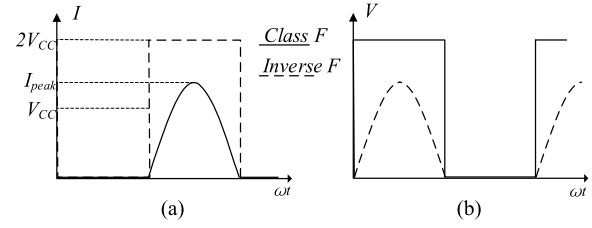


Fig. 1. Ideal class  $F$  and inverse class  $F$  modes. (a) Current waveforms. (b) Voltage waveforms.

the class- $F$  harmonic termination network and its variant, the inverse class- $F$  (class  $F^{-1}$ ), are particularly suitable for optimizing the operating conditions of rectifier diodes. By adjusting these networks, the current and voltage waveforms across the diode are reshaped. For example, in the class  $F^{-1}$  configuration, the voltage exhibits a half-sine waveform while the current maintains a square-waveform; in the class- $F$  configuration, the situation is reversed. This effectively reduces the overlap between voltage and current harmonics, minimizing energy loss in the diode and thus enhancing the overall performance of the rectifier.

To deeply understand the operating principles of a harmonic termination rectifier, we explore it by performing a Fourier series expansion analysis of the waveform shown in Fig. 1. Taking a class  $F^{-1}$  rectifier as an example, in this specific rectification circuit, we assume that all even-order harmonic terminations are open circuits and all odd-order harmonic terminations are short circuits. This harmonic termination strategy is essentially the same as that used in a class  $F^{-1}$  PA.

For a class  $F^{-1}$  rectifier, the voltage across the diode is expressed as a combination of its fundamental frequency component and other higher order harmonics. Based on the harmonic termination conditions, the expression for the fundamental frequency component of the voltage is derived. Typically, this fundamental frequency component is extracted from the original periodic waveform through Fourier analysis, representing the primary frequency part of the signal, while higher order harmonics are frequency components that are integer multiples of the fundamental. Specifically, the fundamental frequency component  $V(\theta)$  in a class  $F^{-1}$  rectifier can be expressed using the following equation:

$$V(\theta) = \begin{cases} 2V(f_0) \sin \theta, & 0 \leq \theta < \pi \\ 0, & \pi \leq \theta < 2\pi. \end{cases} \quad (1)$$

where  $V(f_0)$  is the diode voltage at the fundamental frequency. All even-order harmonics are terminated with an open circuit, while odd-order harmonics are terminated with a short circuit. This implies that odd-order harmonics are suppressed or significantly reduced within the rectifier. The Fourier expansion of the dc voltage  $V_{dc}$  of the waveform is represented by

$$V_{dc} = \frac{2V(f_0)}{\pi}. \quad (2)$$

According to the waveform distribution, the current through the diode is zero during the other half of the cycle, and the

current  $i(\theta)$  is given by the following equation:

$$i(\theta) = \begin{cases} 0, & 0 \leq \theta < \pi \\ 2I_{dc}, & \pi \leq \theta < 2\pi. \end{cases} \quad (3)$$

The Fourier expansion of the current waveform for the dc component is

$$I_{dc} = \frac{\pi}{4} I(f_0) \quad (4)$$

where  $I(f_0)$  is the diode current at fundamental frequency. DC loads are thus derived as

$$\begin{aligned} R_{dc} &= \frac{8V(f_0)}{\pi^2 I(f_0)} \\ &= \frac{8}{\pi^2} R(f_0). \end{aligned} \quad (5)$$

The conversion efficiency in the ideal case can be calculated as

$$\begin{aligned} \eta &= \frac{P_{dc}}{P(f_0)} \\ &= \frac{2V_{dc}I_{dc}}{V(f_0)I(f_0)} \\ &= 1. \end{aligned} \quad (6)$$

Therefore, if the dc load is set to a specific value as given by (6), an idealized half-wave rectifier can achieve 100% efficient conversion from RF power into dc power. Similarly, class  $F$  rectifiers are identical.

To achieve class  $F^{-1}$ , the harmonic control network must provide open-circuit terminations for even-order harmonics and short-circuit terminations for odd-order harmonics. This design ensures that the voltage waveform shapes into a half-sine wave while the current waveform remains a square wave, as shown in Fig. 1, with no overlap between them. This theoretically reaches a 100% power conversion efficiency (PCE).

However, in practical applications, precise control is usually limited to the third harmonic and below. Although theoretically controlling higher order harmonics could slightly improve efficiency, the actual gain is minimal, and the complexity of design and implementation increases significantly. Therefore, for practical class- $F$  power rectifiers employing a finite number of harmonic terminations, the normalized voltage waveform  $V_F(\theta)$  can be simplified and represented as follows: [40]:

$$V_F(\theta) = 1 - \frac{2}{\sqrt{3}} \cos \theta + \frac{1}{3\sqrt{3}} \cos 3\theta. \quad (7)$$

The equations above describe how a class- $F$  PA achieves a theoretical efficiency of approximately 90.7% under maximum power output conditions. The class  $F^{-1}$  PA mode serves as a complementary approach to the class- $F$  mode, with a distinct implementation strategy. In the class  $F^{-1}$  mode, the design focus shifts toward achieving open-circuit terminations for even harmonics and short-circuit terminations for odd harmonics, with a particular emphasis on controlling the peak of the second harmonic. This method reshapes the voltage waveform by altering the harmonic termination conditions, tailoring it to meet different application requirements [41]

$$v_{F^{-1}}(\theta) = 1 + \frac{2}{\sqrt{2}} \cos \theta + \frac{1}{2} \cos 2\theta. \quad (8)$$

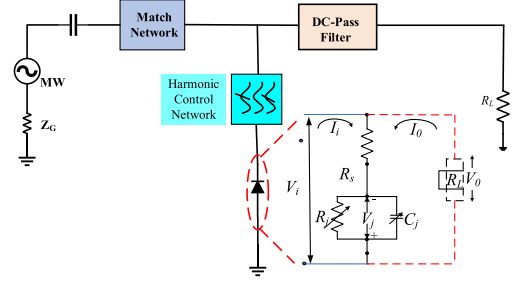


Fig. 2. Diagram of rectifier working principle.

The corresponding current waveform has the form

$$i_{F^{-1}}(\theta) = i_{dc} - i_1 \cos \theta + i_3 \cos 3\theta \quad (9)$$

where  $i_{dc} = 0.37$ ,  $i_1 = 0.43$ , and  $i_3 = 0.06$ .

In this study, we utilized a single parallel-diode class- $F$  rectifier topology for model analysis, as illustrated in Fig. 2. This rectifier primarily consists of a matching network, a harmonic control network, a diode, a dc-pass filter, and a dc load. In addition, the equivalent circuit of the diode used for analysis is also shown in Fig. 2. This equivalent circuit includes a series resistance, a nonlinear junction resistance, and a nonlinear junction capacitance. The value of the nonlinear junction capacitance can be calculated according to the following equation:

$$C_j = C_{j0} \left( 1 - \frac{V_0}{V_{bi}} \right)^{-M} \quad (10)$$

where  $C_{j0}$  is the diode zero bias junction capacitance,  $V_{bi}$  is the diode reverse bias voltage,  $V_0$  is the diode built-in potential, and  $M$  is the diode voltage equalization factor. The nonlinearity of the diode also introduces losses to reduce efficiency, and its efficiency  $\eta_d$  for converting from radio frequency to direct current is expressed as [43]

$$\eta_d = \frac{1}{1 + A + B + C} \times 100\% \quad (11)$$

where

$$\begin{aligned} A &= \frac{R_L}{\pi} \left( 1 + \frac{V_{bi}}{V_0} \right)^2 \\ &\times \left[ \frac{R_s C_j^2 \omega^2}{2} \left( \frac{\theta_{on}}{\cos^2 \theta_{on}} - \tan \theta_{on} \right) \right. \\ &\quad \left. + \frac{1}{R_s} \left\{ \theta_{on} \left( 1 + \frac{1}{2 \cos^2 \theta_{on}} \right) - \frac{3}{2} \tan \theta_{on} \right\} \right] \end{aligned} \quad (12)$$

$$B = \frac{R_s R_L C_j^2 \omega^2}{2\pi} \left( 1 + \frac{V_{bi}}{V_0} \right)^2 \left( \frac{\pi - \theta_{on}}{\cos^2 \theta_{on}} + \tan \theta_{on} \right) \quad (13)$$

$$C = \frac{R_L}{\pi R_s} \left( 1 + \frac{V_{bi}}{V_0} \right) \frac{V_{bi}}{V_0} (\tan \theta_{on} - \theta_{on}) \quad (14)$$

where  $\omega$  is the angular frequency of the input microwave.  $\theta_{on}$ ,  $R_L$ ,  $R_s$ , and  $R_j$  are the forward-bias turn-on angles, dc load, series resistance, and nonlinear junction resistance, respectively. Analysis and model calculations of diode efficiency indicate that, theoretically, a diode with maximum efficiency is achieved by featuring low  $R_s$ , low  $V_{bi}$ , and low  $C_j$ . However, in practical applications, it is highly challenging to meet all

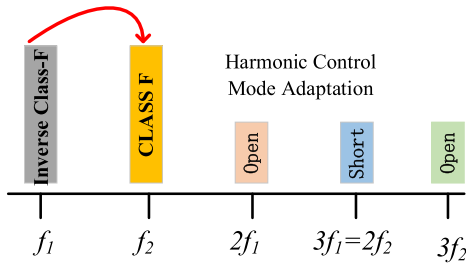


Fig. 3. Principle of adaptive harmonic control.

these ideal parameter values simultaneously due to material properties and manufacturing process limitations. Therefore, in the design process, careful tradeoffs between various diode parameters are required to achieve the performance that approximates the ideal specifications as closely as possible.

### B. Harmonic Control Mode Adaptation

In harmonic suppression rectifiers, microstrip lines are commonly used to tune the diode impedances and impose stringent design requirements on the load impedance at higher order harmonics. It is achieved by adding a parallel open-circuit and a series of short-circuit stubs. In class  $F^{-1}$  design, the second-order current and third-order voltage harmonics are eliminated by designing a  $\lambda/8$  short-circuit line and a  $\lambda/12$  open-circuit microstrip line. In class  $F$  design, a  $\lambda/8$  open-circuit line and a  $\lambda/12$  short-circuit microstrip line are used. This strategy achieves a high rectifying efficiency, but places a strict limit on the rectifier operating frequency as well.

The theory of continuous class  $F$  and class  $F^{-1}$  has demonstrated that strict open-circuit and short-circuit conditions for harmonics are not the only means to maintain high efficiency in PAs. The identical principle applies to rectifiers [41], [42]. By allowing the input voltage of the diode to vary within a certain range, a continuous design space is created to improve the rectifier efficiency. It is widely used in PA design and can also be applied to broadband rectifiers.

Controlling harmonics over a wide frequency range is challenging, and it is particularly difficult to precisely tune the impedance of the third harmonic to open circuit for continuous class  $F$  or short-circuit for class  $F^{-1}$ . However, by analyzing the working principles of both modes, it is found that when the fundamental frequency of the two modes meets certain conditions, both modes can be simultaneously realized within a certain operating bandwidth [44]. Fig. 3 illustrates the principle of mode control conversion.

In a rectifier,  $f_1$  and  $f_2$  represent the starting and cutoff frequencies of the operating bandwidth, respectively, and we assume that  $3f_1 \approx 2f_2$ . In addition, the terminal impedance at  $2f_1$  is close to an open-circuit point, while it is close to a short-circuit point at  $2f_2$ . Therefore, the rectifier operates in a standard class  $F^{-1}$  control mode at  $f_1$  and in a standard class  $F$  control mode at  $f_2$ . By incorporating the continuous design space, the rectifier's bandwidth is significantly extended. It shows how a smooth transition from one mode to another is achieved by precisely controlling the ratio of the fundamental starting and cutoff frequencies, which provides a novel idea for designing broadband rectifiers.

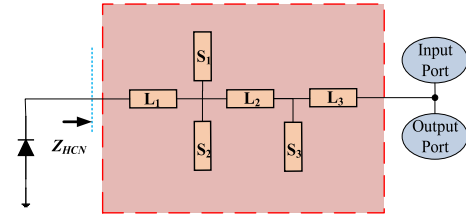


Fig. 4. Low-pass filter as the harmonic control network.

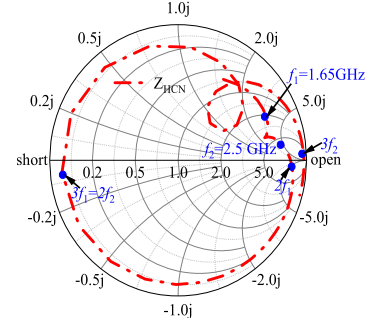


Fig. 5. Harmonic impedance variation diagram.

Appropriate impedance-matching nodes are key enablers for achieving broadband rectifiers. A multistage low-pass topology, implemented using TLs, is an effective method. Designing such a complex network requires network synthesis, which is achieved using real-frequency techniques [44], [45] or methods based on low-pass filter prototypes [46]. Fig. 4 shows a typical three-stage low-pass filter as a harmonic control network. In this network, inductors are realized using high-impedance TLs, while capacitors are implemented using low-impedance open-circuited stubs.

As illustrated in Fig. 5, a well-executed low-pass filter achieves effective impedance compression over the expected frequency range that matches its passband, reducing the difficulty of matching to the signal source. It ensures that harmonic frequencies fall within the stopband of the low-pass filter, with the harmonic impedance located near the edge of the Smith chart. The desired frequency response is achieved by adjusting the characteristic impedances and electrical lengths of the microstrip lines. An electrical length design at  $f_1$  is shown in Table I. Looking at the input impedance of the harmonic control network at the diode in Fig. 4,  $Z_{HCN}$  reaches an open circuit at  $2f_1$ , with  $S_1$  providing a short circuit at  $3f_1$  and  $2f_2$ , respectively; while  $S_2$ , in combination with  $L_1$ , achieves an open circuit of  $Z_{HCN}$  at  $3f_2$ . Harmonic control mode switching for  $f_1$  and  $f_2$  is achieved by impedance transformation of the microstrip line. At the same time, the pure reactance conditions ensure high efficiency in the intermediate frequency band [47].

## III. RECTIFIER DESIGN BASED ON HARMONIC CONTROL NETWORKS

### A. Design of a Broadband Rectifier I

Based on the proposed harmonic terminal mode-switching theory and the harmonic control network, a broadband rectifier was developed to validate the theory. A schematic of the broadband rectifier is shown in Fig. 6, consisting of a matching



TABLE I  
PARAMETERS OF HARMONIC TERMINATION NETWORK

|            | $\theta_{L1}$ | $\theta_{L2}$ | $\theta_{L3}$ | $\theta_{S1}$ | $\theta_{S2}$ | $\theta_{S3}$ |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Unit: deg. | 60            | 20            | 10            | 30            | 20            | 10            |

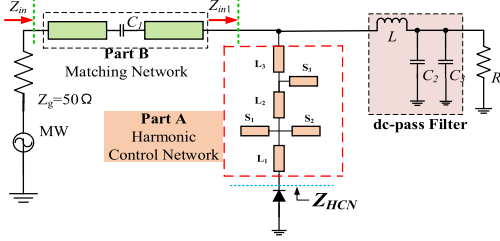


Fig. 6. Schematic of rectifier I based on a harmonic control network.

TABLE II  
SPICE PARAMETERS OF BAT15-03W DIODE

| $B_V$ | $C_{j0}$ | $I_s$ | $R_s$      | $V_{bi}$ |
|-------|----------|-------|------------|----------|
| 6.4 V | 138.5 fF | 74 nA | 5 $\Omega$ | 0.224 V  |

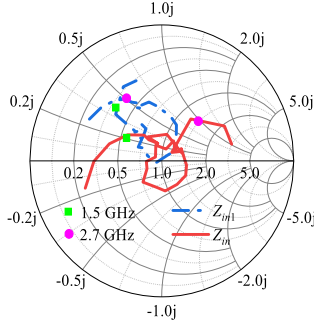


Fig. 7. Impedance matching process of rectifier I at 11 dBm.

network, a harmonic control network, a diode, and a low-pass filter. A BAT15-03W diode with low  $C_j$  and low  $R_s$  was selected for the rectifier design to improve its efficiency. The SPICE parameters of the diode are presented in Table II.

The operating frequencies are set to  $f_1 = 1.65$  GHz and  $f_2 = 2.5$  GHz, and the rectifier was optimized accordingly. First, the output impedance  $Z_{HCN}$  of the diode with the harmonic control network under various frequencies was analyzed. The resulting impedance variations are shown in Fig. 5. At  $f_1 = 1.65$  GHz,  $Z_{HCN}$  at  $2f_1$  reaches the open-circuit region, while  $Z_{HCN}$  at  $3f_1$  reaches the short-circuit region. This configuration meets the impedance requirements for class  $F^{-1}$  harmonic control. At  $f_2 = 2.5$  GHz, the situation is reversed:  $Z_{HCN}$  the at  $2f_2$  is near the short-circuit point, and  $Z_{HCN}$  the at  $3f_2$  is close to the open-circuit point, aligning with the impedance requirements of class- $F$  harmonic control. This demonstrates that the designed harmonic suppression network has a mode-switching function within the operating frequency range, automatically shifts the harmonic impedance state based on the operating frequency, ensuring optimal efficiency at both  $f_1$  and  $f_2$ .

In addition, the introduction of the harmonic control network not only enables higher order harmonic tuning but

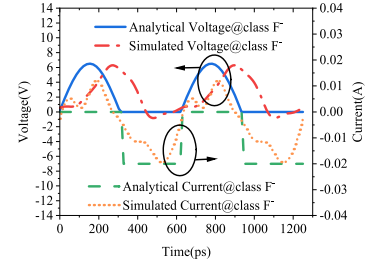


Fig. 8. Diode voltage and current waveforms at 1.65 GHz.

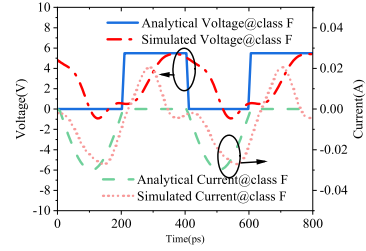


Fig. 9. Diode voltage and current waveforms at 2.5 GHz.

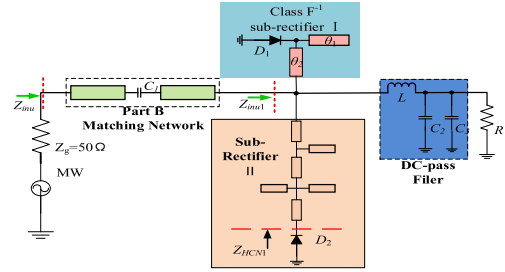


Fig. 10. Schematic of ultrawideband rectifier II.

also effectively compresses the impedance variation of the diode within the operating bandwidth. As shown in Fig. 7, the impedance trajectory after cascading with the harmonic control network is confined near the center of the Smith chart. Consequently, only a simple series microstrip line (Part B) is required to achieve effective broadband impedance matching for the rectifier.

To verify the rectifier's performance under various operating modes, the Advanced Design System (ADS) software was used for voltage and current simulation under large-signal conditions. This approach allows a direct assessment of the rectifier's operation at target frequencies, confirming whether it meets the expected design goals. Fig. 8 shows the voltage and current waveforms at  $f_1 = 1.65$  GHz, closely resembling a standard analytical class  $F^{-1}$  waveform.

Fig. 9 illustrates the waveforms at  $f_2 = 2.5$  GHz, which approximate an ideal class- $F$  operation mode. The waveforms correspond with the results on the Smith chart, validating the feasibility of the proposed mode-switching theory.

### B. Design of an Ultrawideband Rectifier II

To expand on the harmonic control adapting theory, we put more class- $F$  and class  $F^{-1}$  states into a rectifier's operating band, which helps to further extend its bandwidth. Based on the expanded harmonic switching theory, a high-efficiency

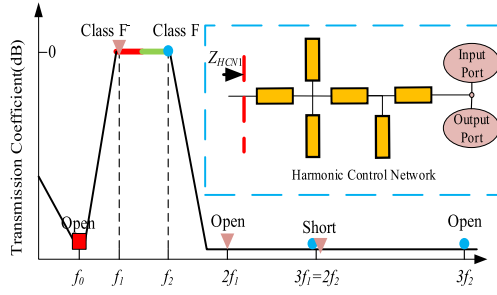


Fig. 11. Frequency response of the novel harmonic control network.

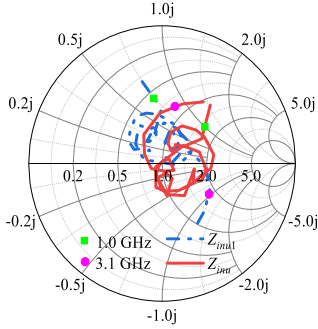


Fig. 12. Impedance matching process of ultrawideband rectifier II at 12 dBm.

ultrawideband rectifier was developed and designed, with its schematic shown in Fig. 10. Rectifier II contains two diodes, each of which combines with an HCN to form a subrectifier. Unlike rectifier I, an additional harmonic terminal subrectifier with a center frequency of  $f_0 \approx 2/3f_1$  was parallelly added, dedicated to collecting low-frequency microwaves.

To ensure both subrectifiers maintain high conversion efficiency, the harmonic control network needs to be further optimized. In addition to meeting the requirements for harmonic switching, the impedance  $Z_{HCN1}$  of the control network must also exhibit a high-impedance state at  $f_0$ , reducing the inflow of microwave around  $f_0$ . The frequency response of the improved harmonic control network is shown in Fig. 11. Subrectifier I operates in the class  $F^{-1}$  mode. For the design of the terminal network, the electrical lengths are set to  $\theta_1 = 1/12$  and  $\theta_2 = 1/4$ . The microstrip line with electric length  $\theta_1 = 1/12$  exhibits a short circuit at  $3f_0$ .

Meanwhile, the operating band of subrectifier II includes  $2f_0$ , which not only provides an effective open circuit for subrectifier I but also prevents high-frequency microwaves from entering subrectifier I. The two subrectifiers work in coordination, and each of them operates efficiently in its respective frequency band.

Similar to rectifier I, the waveforms exhibit the typical voltage and current waveforms of class  $F^{-1}$  and class-F modes, respectively. Similarly, as shown in Fig. 12, the impedance of the two subrectifiers cascaded with harmonic control networks connected in parallel is still confined near the center of the

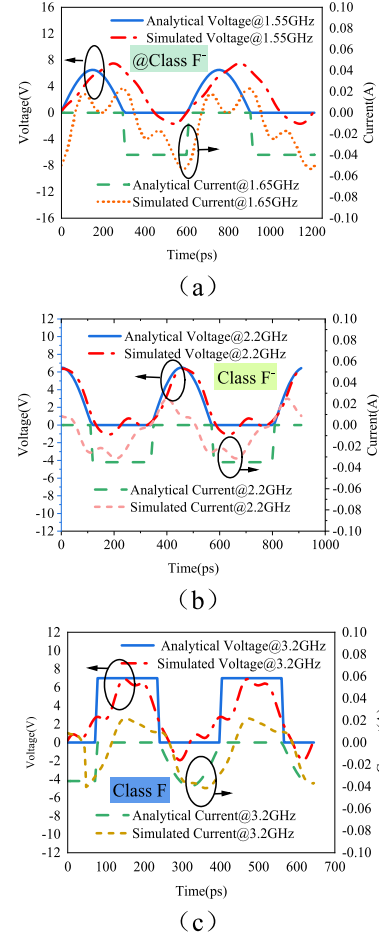


Fig. 13. Diode voltage and current waveforms at (a) 1.55, (b) 2.2, and (c) 3.2 GHz.

Smith chart. Even when the operating bandwidth extends to the ultrawideband range, effective impedance matching can still be achieved using only a simple series microstrip line.

When the two subrectifiers operate independently, they meet the impedance requirements for harmonic termination. The working frequencies  $f_0, f_1$ , and  $f_2$  are set to 1.55, 2.2, and 3.2 GHz, respectively. The voltage and current waveforms are analyzed, as shown in Fig. 13. Fig. 13(a) illustrates the segmented voltage and current waveforms of diode  $D_1$  at  $f_0$ , where the voltage resembles a half-sine wave and the current approximates a square wave, demonstrating class  $F^{-1}$  characteristics. Fig. 13(b) and (c) shows the voltage and current waveforms of diode  $D_2$  at  $f_1$  and  $f_2$ , respectively. Similar to rectifier I, the waveforms exhibit the typical voltage and current waveforms of class  $F^{-1}$  and class-F modes, respectively.

By analyzing the circuit topology after the parallel connection of the two subrectifiers, the results are shown in Figs. 14 and 15. Fig. 14 illustrates the S-parameters of the ultrawideband rectifier where different subrectifiers are active. When only diode  $D_1$  is operating, the rectifier achieves good matching around  $f_0$ . When only diode  $D_2$  is active, the rectifier

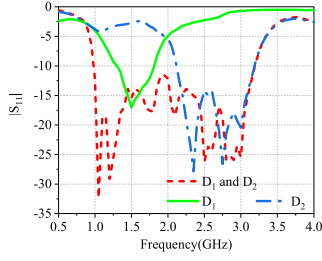


Fig. 14. S-parameters of the ultrawideband rectifier at three operating modes: only  $D_1$ , only  $D_2$ , and both  $D_1$  and  $D_2$ .

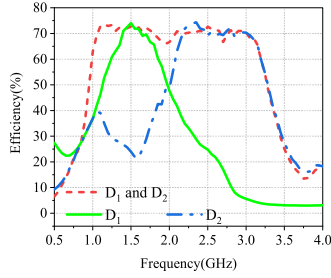


Fig. 15. Efficiency of the ultrawideband rectifier with three operating modes.

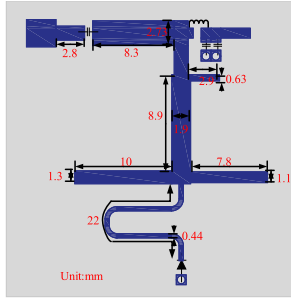


Fig. 16. Layout of the broadband rectifier I.

achieves good matching from 2 to 3.3 GHz. When the two subrectifiers are connected in parallel, the rectifier achieves good matching over a broader frequency range from 1 to 3.3 GHz. The simulated efficiency is shown in Fig. 15. These results closely correspond with the simulated  $|S_{11}|$  curve. When only diode  $D_1$  operates, the peak efficiency at 1.5 GHz exceeds 70%, while diode  $D_2$  achieves over 70% efficiency within the 2–3.3-GHz range. A notable result is that when both diodes operate simultaneously, their operating characteristics combine effectively, extending the bandwidth and achieving a rectification efficiency of over 70% across 1–3.1 GHz, with a fractional bandwidth exceeding 100%. This validates the scalability and effectiveness of the harmonic control theory, demonstrating the capability to achieve high efficiency across an ultrawideband frequency range while reducing the dependency of harmonic termination on specific frequencies.

#### IV. MEASUREMENT RESULTS AND ANALYSES

##### A. Analyses of Broadband Rectifier I

To verify the feasibility of the design theory, two rectifiers were fabricated based on the proposed design. The rectifiers

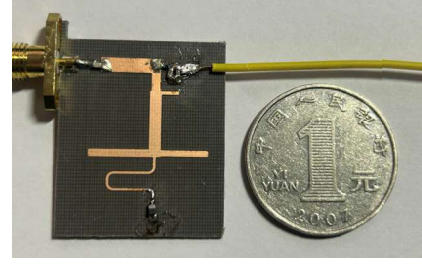


Fig. 17. Photograph of the broadband rectifier I.

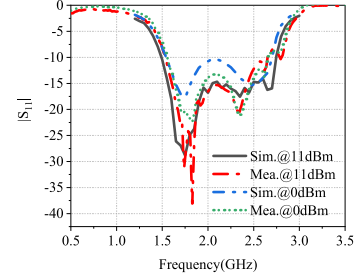


Fig. 18. S-parameter of the broadband rectifier I.

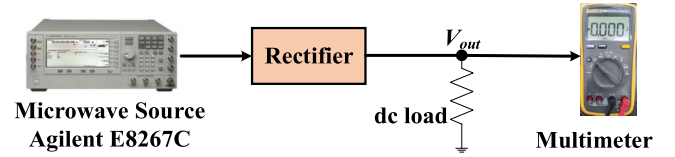


Fig. 19. Diagram of the measurement system.

were implemented on an F4B-M dielectric substrate, with a relative permittivity ( $\epsilon_r$ ) of 2.65 and a loss tangent ( $\tan \delta$ ) of 0.0012. The substrate thickness is 1 mm. First, the layout and physical prototype of the broadband rectifier I are shown in Figs. 16 and 17. The specific design parameters are detailed in Fig. 16. The lumped components used in the rectifier I include  $C_1$  (3 pF),  $C_2$  (47 pF),  $C_3$  (68 pF), and  $L$  (33 nH).

First, the S-parameters of the fabricated rectifier were measured using a vector network analyzer (Agilent N5230A). The measurement results are shown in Fig. 18. It is observed that the simulation and measurement results align closely. Under microwave power levels of 0 and 11 dBm,  $|S_{11}|$  of the rectifier I remains below -10 dB across the frequency range of 1.5–2.7 GHz, indicating good impedance matching.

Then, the PCE of rectifier I was measured. Fig. 19 illustrates the measurement setup for rectifier efficiency. A microwave signal generator (Agilent E8267C) was used to provide microwave power to the rectifier. A decade resistance box served as the output load to receive the dc power, and a voltmeter was used to measure the dc output voltage. The microwave to dc conversion efficiency, denoted as  $\eta_{MW-dc}$ , is

$$\eta_{MW-dc} = \frac{P_{dc}}{P_{IN}} \times 100\%$$

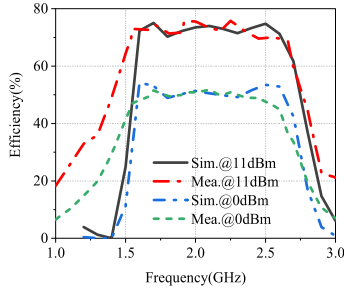


Fig. 20. Efficiency of the broadband rectifier I at different power levels.

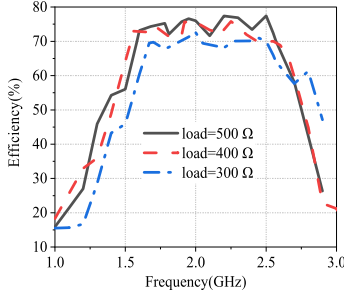


Fig. 21. Efficiency curves of broadband rectifier I at different dc loads.

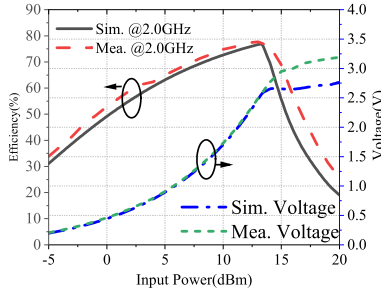


Fig. 22. Optimum efficiency and voltage for broadband rectifier I at 2 GHz.

$$= \frac{V_{\text{OUT}}^2}{R_{\text{dc}}} \times \frac{1}{P_{\text{IN}}} \times 100\% \quad (15)$$

where  $V_{\text{OUT}}$  is the output dc voltage, and  $P_{\text{IN}}$  is the input microwave power.

The simulated and measured rectification efficiencies under different input power levels are shown in Fig. 20, with a dc load of 400  $\Omega$ . The simulated efficiency closely matches the measured results. Under a microwave input power of 11 dBm, the rectification efficiency exceeds 70% across the frequency band of 1.54–2.66 GHz, achieving a fractional bandwidth of 53.3%. When the input power decreases to 0 dBm, the rectification efficiency remains near 50%.

Measurement results for different output loads are shown in Fig. 21. Increasing the load resistance improves the rectification efficiency but reduces the bandwidth to some extent. The peak efficiency frequency of rectifier I is also measured, as shown in Fig. 22. Rectifier I achieves a peak efficiency of 77.7% at the center frequency of 2 GHz under an input power of 13 dBm. In addition, a slight frequency shift is observed between the simulated and measured results, likely due to manufacturing tolerances

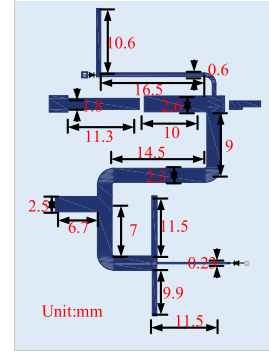


Fig. 23. Layout of the ultrawideband rectifier II.

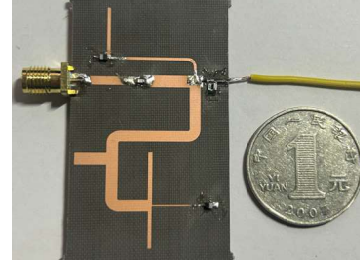


Fig. 24. Photograph of the ultrawideband rectifier II.

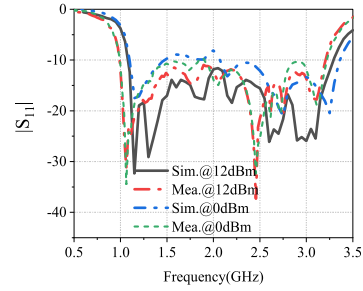


Fig. 25. S-parameter of the ultrawideband rectifier II.

and discrepancies between the diode model and the actual device.

### B. Analyses of Ultrawideband Rectifier II

Ultrawideband rectifier II was also fabricated and measured to validate the harmonic control extension. Fig. 23 shows the design layout and design parameters of the ultrawideband rectifier II. Fig. 24 presents the actual photograph of the rectifier, which was fabricated using the same dielectric substrate. The lumped components used in rectifier II are as follows:  $C_1$  (3.9 pF),  $C_2$  (47 pF),  $C_3$  (68 pF), and  $L$  (30 nH).

The S-parameters of the ultrawideband rectifier II were measured under microwave power levels of 12 and 0 dBm, as shown in Fig. 25. Both simulation and measurement results show that the ultrawideband rectifier II maintains good impedance matching with  $|S_{11}|$  below  $-10$  dB across different input power levels. Compared with rectifier I, the matching bandwidth of this rectifier extends from 1 to 3.2 GHz, effectively doubling the operating bandwidth. The extension of



TABLE III  
COMPARISON WITH WIDEBAND RECTIFIERS

| Ref                 | PCE > 70% (GHz)  | Fraction Bandwidth | Power (dBm) | PCE > 40% at 0 dBm (GHz) | Fraction Bandwidth | Maximum Efficiency | Size (mm <sup>2</sup> ) |
|---------------------|------------------|--------------------|-------------|--------------------------|--------------------|--------------------|-------------------------|
| [22]                | 1.47-1.77        | 18.50%             | 10          | N/A                      | N/A                | 80.00%             | 71×20                   |
| [26]                | 2.08-2.58        | 21.50%             | 15.5        | 2.03-2.75                | 30%                | 80.80%             | 126×68                  |
| [30]                | 2.1-3.3          | 44.40%             | 14          | N/A                      | N/A                | 76.30%             | 31×18                   |
| [27]                | 0.57-0.90        | 44.90%             | 12.8        | N/A                      | N/A                | 76.00%             | 60×88                   |
| [32]                | 1.77-2.85        | 46.8%              | 11.5        | 1.73-2.93*               | 51.5%              | 82.3%              | 26×26                   |
| [31]                | 1.6-2.8          | 54.50%             | 11          | 1.7-2.8                  | 49%                | 78.63%             | 37×10                   |
| [21]                | 1.75-3.55        | 67.9%              | 10          | N/A                      | N/A                | N/A                | 25×25                   |
| [34]                | 1-2.7            | 91.9%              | 15          | N/A                      | N/A                | 78.2%              | 38×15                   |
| [23]                | 1.1-3.1          | 95.2%              | 30          | N/A                      | N/A                | 75.8%              | 30×30*                  |
| <b>Rectifier I</b>  | <b>1.54-2.66</b> | <b>53.3%</b>       | <b>11</b>   | <b>1.52-2.76</b>         | <b>57.9%</b>       | <b>77.7%</b>       | <b>32×25</b>            |
| <b>Rectifier II</b> | <b>1-3.1</b>     | <b>102.4%</b>      | <b>12</b>   | <b>1-3.2</b>             | <b>104.8%</b>      | <b>77.1%</b>       | <b>51×33</b>            |

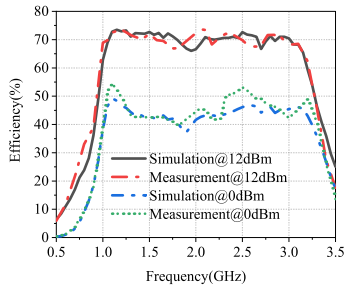


Fig. 26. Efficiency of the ultrawideband rectifier II at different power levels.

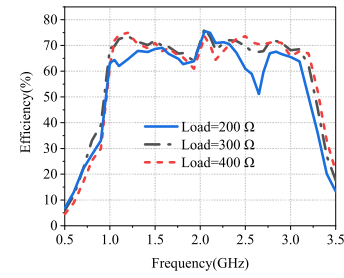


Fig. 28. Efficiency of the ultrawideband rectifier II at different dc loads.

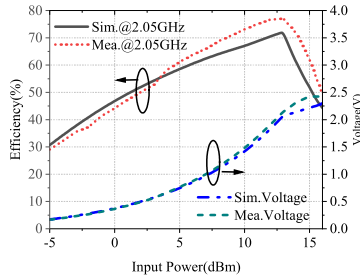


Fig. 27. Efficiency and dc output voltage of the ultrawideband rectifier II at 2.05 GHz.

the harmonic control network has significantly enhanced the operating bandwidth of the rectifier.

The PCE of the ultrawideband rectifier under different conditions was also measured. Fig. 26 illustrates the conversion efficiency of rectifier II at 0 and 12 dBm across different frequencies, with a dc load of  $R_{dc} = 300 \Omega$ . It shows that the rectifier achieves an efficiency greater than 70% in the frequency range from 1 to 3.1 GHz, with a bandwidth reaching three times the center frequency and a fractional bandwidth exceeding 100%. When the input power is reduced to 0 dBm, the efficiency remains over 40% in the frequency range of 1–3.25 GHz, further extending the operating bandwidth. The highest rectification efficiency is also achieved at the center frequency of 2.05 GHz, with an efficiency of 77.1% at 13 dBm, as shown in Fig. 27. The

deviation from the simulation is primarily due to the increase in the operating frequency band, as the simulation diode model is inaccurate, leading to a slight mismatch between the simulated and actual optimal working frequency of rectifier II. The conversion efficiency under different dc loads is shown in Fig. 28.

Table III compares the two proposed rectifiers in this work with previous designs. The rectifier proposed in [31] achieves a wide bandwidth with nice performance. Rectifier I exhibits higher efficiency across the operating band, particularly at low input power levels (0 dBm). The thresholds of efficiency >70% or >40% are used for clearer comparison. The harmonic control network not only extends the bandwidth but also contributes to improving the overall efficiency. The ultrawideband rectifier II has a significantly broader operating bandwidth than other designs, being the only one to achieve a three-times fractional bandwidth. This validates the advancement and effectiveness of our design theory, significantly extending the operational frequency range of the rectifier.

## V. CONCLUSION

This article presents a design method for ultrawideband rectifiers based on frequency-adaptive harmonic control, which not only improves the rectifier efficiency but also extends its bandwidth. This article begins with an analysis of how harmonic terminal control enhances conversion efficiency, followed by an examination of the working principle of

the harmonic control network and proposed design improvements. A broadband rectifier and an extended ultrawideband rectifier were fabricated and tested to validate the theory. The high-efficiency working bandwidth exceeding 100% demonstrates the advanced and effective nature of the proposed theory.

The introduction of this theory significantly enhances the operational bandwidth of a rectifier, advancing the development of wireless power transfer and spatial EH technologies. The high-efficiency ultrawideband rectifier effectively improves system stability and is suitable for applications in powering a sensor network or low-power electronic devices.

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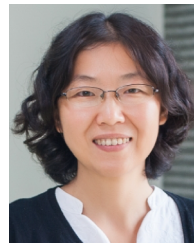
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